easyep.py

`src/moe_pruning/scoring/easyep.py · 96 lines · module moe_pruning.scoring.easyep`

What it does

The EASY-EP formulas, verified against the official implementation: output-aware importance c = g * ||E(x)||, token contribution s = max(1 - cos(h, h + y), 0), and expert importance I = sum over tokens of c * s.

How it works

- `token_contribution()` and `output_aware_importance()` compute the two per-token quantities.
- `rank_experts()` orders experts from least to most important; unobserved experts are placed last.
- `select_lowest()` picks the experts to replace and refuses to remove all experts of a layer.

Notes

- The module docstring lists the differences to the DeepSeek setting of the paper (top-1/top-k routing, no shared experts, capacity-dropped tokens, padding).

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`token_contribution(residual, moe_output)`	s_t = max(1 - cos(h_t, h_t + y_t), 0), computed in float32. Shapes [..., D] -> [...].
`output_aware_importance(gate, expert_output)`	c_{e,t} = g_{e,t} * ||E_e(x_t)||_2. Shapes gate [...], expert_output [..., D] -> [...].
`rank_experts(importance, observed)`	Expert ids ordered from least to most important.
`select_lowest(importance, observed, num_to_select, allow_unobserved)`	Pick the `num_to_select` least important experts of one layer.

Connections

Used by: `src/moe_pruning/pruning/expert_selection.py`, `src/moe_pruning/scoring/statistics.py`